

Yingheng Wang

CS PhD Candidate (AI/ML) | Cornell University

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Education

Cornell University

Aug. 2022 - present

Committee: [Carla P. Gomes](#) (Chair), [Volodymyr Kuleshov](#), [Christopher De Sa](#)

Ithaca, NY

PhD, Computer Science (AI/ML) | GPA: 3.91/4.0

Cornell University

Aug. 2022 - Dec. 2025

Ithaca, NY

MS, Computer Science | GPA: 4.0/4.0

Johns Hopkins University

Aug. 2021 - Jun. 2022

Baltimore, MD

MS, Biomedical Engineering | GPA: 4.0/4.0

Tsinghua University

Sep. 2020 - Jun. 2022

Beijing, China

MS, Electronic Engineering | GPA: 4.0/4.0

Tongji University

Sep. 2016 - Jun. 2020

Shanghai, China

BS, Electrical and Computer Engineering | GPA: 95/100 (Ranking: 1/110)

Research Interest

My research interest lies in knowledge-centric AI with an emphasis on scientific reasoning. More specifically, I research AI reasoning (probabilistic inference/reinforcement learning/search/optimization/sampling), generative modeling (large language model/diffusion model/flow matching/optimal transport), and representation learning (pre-training/self-supervised learning/unsupervised learning/semi-supervised learning/geometric learning).

Research Impact

I have published research in leading AI/ML and science venues, including ICML, NeurIPS, ICLR, AAAI, TMLR, Nature, and Nature Machine Intelligence. My work has garnered 1,700+ citations, with several papers receiving Best Paper Awards, Featured Paper Awards, and winning ML Challenges. My intern projects have also led to impactful patents adopted by big AI companies.

Publications (*Equal contribution. †Correspondence.)

Selected:

- [1] **Yingheng Wang**, Tao Yu, Shufeng Kong, Francesco Ricci, John M. Gregoire, Carla P. Gomes. “Deep Scientific Reasoning under Physical Constraints: Structure-Aware Spectrum Prediction for Electronic Density of States”. **ICML**, 2026 [Under Review](#)
- [2] **Yingheng Wang**, Can Chen, Karla-Luise Herpoldt, Sukanya Sasmal, Tommi Jaakkola, Marcus D. Collins, Shang Shang. “Iterative Multi-Objective Policy Optimization for Antibody Sequence Design”. **ICML**, 2026 [Under Review](#)
- [3] Zhongdi Qu, **Yingheng Wang**, Utku Umur Acikalin, Aaron M. Ferber, Goncalo J. Gouveia, Brandon Bills, Guohui Li, Joshua Kline, Sunandini Yedla, Xiao Wang, Frank C. Schroeder, Carla P. Gomes. “Unsupervised Combinatorial Probabilistic Reasoning: Probabilistic Coin Change Problem”. **AAAI**, 2026 [\[paper\]](#)
- [4] Hongzheng Chen*, **Yingheng Wang***, Yaohui Cai*, Hins Hu*, Jiajie Li*, Shirley Huang, Chenhui Deng,

Rongjian Liang, Shufeng Kong, Haoxing Ren, Samitha Samaranayake, Carla P. Gomes, Zhiru Zhang. "Heuri-Gym: An Agentic Benchmark for LLM-Crafted Heuristics in Combinatorial Optimization". **ICLR**, 2026 [\[paper\]](#) [\[code\]](#)

[5] **Yingheng Wang**, Zichen Wang, Gil Sadeh, Luca Zancato, Alessandro Achille, George Karypis, Huzefa Rangwala. "LC-PLM: Long-context Protein Language Modeling Using Bidirectional Mamba with Shared Projection Layers". **TMLR**, 2025 [\[paper\]](#) [\[code\]](#) 🏆 **Featured Certification Award**

[6] **Yingheng Wang***, Top Wasu Piriyaikulij*, Volodymyr Kuleshov. "Diffusion Variational Inference: Diffusion Models as Expressive Variational Posteriors". **AAAI**, 2025 [\[paper\]](#) [\[code\]](#)

[7] **Yingheng Wang**, Shufeng Kong, John M. Gregoire, Carla P. Gomes. "Conformal Crystal Graph Transformer with Robust Encoding of Periodic Invariance". **AAAI**, 2024 [\[paper\]](#) [\[code\]](#)

[8] Junjie Yin, Jiahao Dong, **Yingheng Wang**, Christopher De Sa, Volodymyr Kuleshov. "ModuLoRA: Fine-tuning 2-Bit LLMs on Consumer GPUs by Integrating with Modular Quantizers". **TMLR**, 2024 [\[paper\]](#) [\[code\]](#) 🏆 **Featured Certification Award**

[9] **Yingheng Wang**, Yair Schiff, Aaron Gokaslan, Weishen Pan, Fei Wang, Christopher De Sa, Volodymyr Kuleshov. "InfoDiffusion: Representation Learning Using Information Maximizing Diffusion Models". **ICML**, 2023 [\[paper\]](#) [\[code\]](#)

[10] Yuanqi Du*, **Yingheng Wang***, Yining Huang, Jianan Canal Li, Yanqiao Zhu, Tian Xie, Chenru Duan, John M. Gregoire, Carla P. Gomes. "M²Hub: Unlocking the Potential of Machine Learning for Materials Discovery". **NeurIPS**, 2023 [\[paper\]](#) [\[code\]](#)

[11] He S. Yang*, Weishen Pan*, **Yingheng Wang***, Mark A. Zaydman, Nicholas C. Spies, Zhen Zhao, Theresa A. Guise, Qing H. Meng, Fei Wang. "Generalizability of a Machine Learning Model for Improving Utilization of Parathyroid Hormone-Related Peptide Testing Across Multiple Clinical Centers". **Clinical Chemistry**, 2023 [\[paper\]](#) [\[code\]](#) 🏆 **The winner solution of the ML challenge (PTHrP Result Prediction) at AACC 2022**

[12] Zhongyu Huang*, **Yingheng Wang***, Chaozhuo Li, Huiguang He. "Going Deeper into Permutation-Sensitive Graph Neural Networks". **ICML**, 2022 [\[paper\]](#) [\[code\]](#)

[13] **Yingheng Wang**, Yaosen Min, Xin Chen, Ji Wu. "Multi-view Graph Contrastive Representation Learning for Drug-drug Interaction Prediction". **The Web Conf (WWW)**, 2021 [\[paper\]](#) [\[code\]](#) 🏆 **Best Paper Award in AAAI GraphDL 2021**

[14] **Yingheng Wang**, Yaosen Min, Erzhuo Shao, Ji Wu. "Molecular Graph Contrastive Learning with Parameterized Explainable Augmentations". **BIBM**, 2021 [\[paper\]](#)

[15] **Yingheng Wang**, Xin Chen, Yaosen Min, Ji Wu. "MolCloze: A Unified Self-supervised Molecular Structure Pre-training Model for Chemical Property Prediction". **BIBM**, 2021 [\[paper\]](#)

Others:

[16] Center for AI Safety, Scale AI, HLE Contributors Consortium (incl. **Yingheng Wang**). "A benchmark of expert-level academic questions to assess AI capabilities". **Nature**, 2026 [\[paper\]](#) [\[code\]](#)

[17] Jiarui Liu, Kaustubh Dhole, **Yingheng Wang**, Haoyang Wen, Sarah Zhang, Haitao Mao, Gaotang Li, Neeraj Varshney, Jingguo Liu, Xiaoman Pan. "Toward Honest Language Models for Deductive Reasoning". **ICML**, 2026 [\[paper\]](#) **Under Review**

[18] Shufeng Kong, Xingru Yang, Yuanyuan Wei, Zijie Wang, Sidie Huang, Hao Tang, Jiuqi Qin, Shuting Lan, **Yingheng Wang**, Junwen Bai, Zhuangbin Chen, Zibin Zheng, Caihua Liu, Liang Hao. "MTCMB: A Multi-Task Benchmark for Evaluating LLMs on Knowledge, Reasoning, and Safety in Traditional Chinese Medicine". **npj Digital Medicine**, 2026 [\[paper\]](#) **Under Review**

[19] Shufeng Kong, Zijie Wang, Nuan Cui, Tang Hao, Meng Yihan, Yuan Yuan Wei, Chenfei Fan, **Yingheng Wang**, Zhuo Cai, Yaonan Wang, Yulong Zhang, Yuzheng Li, Zibin Zheng, Caihua Liu, Liang Hao. "MIRNet: Integrating Constrained Graph-Based Reasoning with Pre-training for Diagnostic Medical Imag-

ing". [AAAI](#), 2026 [\[paper\]](#)

[20] Taoyuze Lv, Alexander Chen, Fengyu Xie, Jeffrey Meng, **Yingheng Wang**, Bram Hoex, Zhicheng Zhong, Tong Xie. "AtomWorld: A Benchmark For Evaluating Spatial Reasoning In Large Language Models On Crystalline Materials". [NeurIPS AI4Mat](#), 2025 [\[paper\]](#)

[21] Christian K. Belardi, Chia-Hao Lee, **Yingheng Wang**, Justin Lovelace, Kilian Q. Weinberger, David A. Muller, Carla P. Gomes. "Improving Electron Ptychography with a Generative Prior". [ICCV](#), 2025 [\[paper\]](#)

[22] Chia-Hao Lee, Christian K. Belardi, **Yingheng Wang**, Justin Lovelace, Kilian Q. Weinberger, David A. Muller, Carla P. Gomes. "3D Ptychographic Inverse Imaging with Generative Diffusion Models". [Microscopy and Microanalysis](#), 2025 [\[paper\]](#)

[23] Yuwei Wan, Yuqi An, Dongzhan Zhou, Jiahao Dong, Chunyu Kit, Wenjie Zhang, Bram Hoex, Tong Xie[†], **Yingheng Wang**[†]. "MatFusion: A Multi-Modal Framework Bridging LLMs and Structural Embeddings for Experimental Materials Property Prediction". [ICLR AI4Mat](#), 2025 [\[paper\]](#)

[24] Yuanqi Du*, Arian R Jamasb*, Jeff Guo*, Tianfan Fu, Charles Harris, **Yingheng Wang**, Chenru Duan, Pietro Liò, Philippe Schwaller, Tom L Blundell. "Machine learning-aided generative molecular design". [Nature Machine Intelligence](#), 2024 [\[paper\]](#)

[25] Brendan H. Rappazzo, **Yingheng Wang**, Aaron M. Ferber, Carla P. Gomes. "GEM-RAG: Graphical Eigen Memories For Retrieval Augmented Generation". [arXiv](#), 2024 [\[paper\]](#)

[26] Dingshuo Chen, Shuchen Xue, Liuji Chen, **Yingheng Wang**, Qiang Liu, Shu Wu, Zhi-Ming Ma, Liang Wang. "Graffe: Graph Representation Learning Enabled via Diffusion Probabilistic Models". [TKDE](#), 2025 [\[paper\]](#) [Under Review](#)

[27] Zhongyu Huang, **Yingheng Wang**, Chaozhuo Li, Huiguang He. "Growing Like a Tree: Finding Trunks From Graph Skeleton Trees". [IEEE TPAMI](#), 2024 [\[paper\]](#) [\[code\]](#)

[28] Tong Zhao, Wei Jin, Yozen Liu, **Yingheng Wang**, Gang Liu, Stephan Gunnemann, Neil Shah, Meng Jiang. "Graph Data Augmentation for Graph Machine Learning: A Survey". [IEEE Data Engineering Bulletin](#), 2023 [\[paper\]](#)

[29] Dongsheng Luo, Wei Cheng, **Yingheng Wang**, Dongkuan Xu, Jingchao Ni, Wenchao Yu, Haifeng Chen, Xiang Zhang. "Information-Aware Time Series Meta-Contrastive Learning". [AAAI](#), 2023 [\[paper\]](#) [\[code\]](#)

[30] Zhongyu Huang, Changde Du, **Yingheng Wang**, Huiguang He. "Graph-Enhanced Emotion Neural Decoding". [IEEE TMI](#), 2023 [\[paper\]](#)

[31] Junwen Bai, Yuanqi Du, **Yingheng Wang**, Shufeng Kong, John M. Gregoire, Carla P. Gomes. "Xtal2DoS: Attention-based Crystal to Sequence Learning for Density of States Prediction". [NeurIPS AI4Science](#), 2022 [\[paper\]](#) [\[code\]](#)

[32] Zhongyu Huang, Changde Du, **Yingheng Wang**, Huiguang He. "Graph Emotion Decoding from Visually Evoked Neural Responses". [MICCAI](#), 2022 [\[paper\]](#)

[33] Erzhao Shao*, Jie Feng*, **Yingheng Wang**, Tong Xia, Yong Li. "One-shot Transfer Learning for Population Mapping". [CIKM](#), 2021 [\[paper\]](#)

[34] Y. Min, Y. Wei, P. Wang, N. Wu, S. Bauer, S. Zheng, Y. Shi, **Yingheng Wang**, X. Wang, D. Zhao, J. Wu, J. Zeng. "From Static to Dynamic Structures: Improving Binding Affinity Prediction with Graph-Based Deep Learning". [Advanced Science](#), 2024 [\[paper\]](#)

[35] B. Yang, T. Meng, X. Wang, J. Li, S. Zhao, **Yingheng Wang**, S. Yi, Y. Zhou, Y. Zhang, L. Li, L. Guo. "CAT Bridge: An Efficient Toolkit for Compound-Transcript Association Mining from Multi-Omics Data". [Giga-Science](#), 2024 [\[paper\]](#)

[36] C. Li, T. He, **Yingheng Wang**, et al.. "Pipeline Image Haze Removal System Using Dark Channel Prior on Cloud Processing Platform". [International Journal of Computational Science and Engineering](#), 2020 [\[paper\]](#)

[37] T. Teng, **Yingheng Wang**, et al.. "Mathematical Modeling and Simulation on the Stimulation Interactions

in Coalbed Methane Thermal Recovery". [Processes](#), 2019 [\[paper\]](#)

Awards & Honors

- Featured Certification Award (invited to present at ICLR 2025), TMLR 2025
- Google AI Academic Program GCP Credit Award (\$10,000) 2024
- Featured Certification Award (invited to present at ICLR 2024), TMLR 2024
- The Winner of the ML Challenge (PTHrP Result Prediction), AACC 2022
- The Participation Scholarship, ICML 2023, 2022
- Presidential Life Science Fellowship, Cornell University 2022
- Best Paper Award, The Workshop on Deep Learning on Graph, AAAI 2021
- Graduate Scholarship, Tsinghua University 2021
- Outstanding Graduate Award (top 0.2%), China Government 2020
- Best Undergraduate Thesis (top 0.2%), China Government 2020
- Outstanding Undergraduate Researcher Award (top 0.2%) 2019
- Meritorious Winner, in Mathematical Contest In Modeling (MCM) 2019
- China National Scholarship (top 0.2%) 2018
- First-Prize Scholarship (top 1%) 2017

Work Experience

Grand Challenge, Amazon NYC, NY, 2024
Applied Scientist Intern policy optimization for antibody sequence design

- Managers: [Marcus D. Collins](#), [Shang Shang](#)
- Mentors: [Tommi S. Jaakkola](#), [Karla-Luise Herpoldt](#), [Can Chen](#)

AWS AI Research & Education, Amazon NYC, NY, 2024
Applied Scientist Intern long-context protein modeling with graphs

- Managers: [Zichen Wang](#), [Huzefa Rangwala](#)
- Mentors: [George Karypis](#), [Alessandro Achille](#), [Luca Zancato](#)

NEC Laboratories America Princeton, NJ, 2021
Research Intern times-series contrastive learning

- Manager: [Wei Cheng](#)
- Mentor: [Dongsheng Luo](#)

University of Washington Seattle, WA, 2021
Research Assistant drug relation mining with multi-modal data

- Advisor: [Sheng Wang](#)

Johns Hopkins University Baltimore, MD, 2021-2022
Research Assistant GNNs for brain network analysis

- Advisor: [Joshua T. Vogelstein](#)
- Mentor: [Sambit Panda](#)

Microsoft Research Asia Beijing, 2020-2022
Research Assistant expressivity of GNNs & long-range GNNs

- Manager: [Chaozhao Li](#)

Tsinghua University
Research Assistant

- Advisor: [Ji Wu](#)
- Mentor: [Yaosen Min](#)

Beijing, 2020-2022
machine learning for drug discovery

Penn State University
Research Assistant

State College, PA, 2019
multi-view graph clustering

- Mentors: [Dongsheng Luo](#), [Dongkuan Xu](#)

Professional Service

Area Chair: NeurIPS (AI4Science)

Conference Reviewer: ICML, NeurIPS, ICLR, AAAI, AISTATS, UAI, CVPR, ICCV, ECCV

Journal Reviewer: IEEE Transactions on Knowledge and Data Engineering (TKDE), Bioinformatics, Journal of Biomedical Informatics, Smart Health, Acta Automatica Sinica

Workshop Reviewer/PC: NeurIPS (Math-AI, AI4Science, GenBio, GLFrontiers), ICLR (AGI), KDD (Health Day)

Others: Co-organizer of Learning on Graph Conference (LoG 2025, 2024, 2023) local meetup, Cornell CS PhD Admission Committee Member (2025, 2024), Cornell CS Visit Day CZar

Invited Talk

- **Grand Challenge @ Amazon, AbMPO**, hosted by Marcus D. Collins 2025
- **Machine Learning Day @ ASU, LC-PLM**, hosted by Yixuan He 2025
- **Multi-omics Reading Group @ MILA, LC-PLM**, hosted by Xinyu Yuan 2025
- **AI4Protein Seminar, LC-PLM**, hosted by Jun Li 2024
- **AWS AI Research & Education @ Amazon, LC-PLM**, hosted by Zichen Wang 2024
- **Toyota Research Institute, CrystalFormer & Xtal2DoS**, hosted by Santosh Suram 2023
- **Weill Cornell Medicine, InfoDiffusion**, hosted by Fei Wang 2023
- **Florida International University, InfoDiffusion**, hosted by Dongsheng Luo 2023
- **American Association for Clinical Chemistry, Championship Solution for the PTHrP Result Prediction Challenge**, hosted by Nicolas Spies 2022
- **AAAI GraphDL 2021, MIRACLE**, hosted by Lingfei Wu 2021

Teaching

- **CS 2770: Computational Sustainability**, Cornell University 2026
Undergraduate Level, *Teaching Assistant*
- **CS 6703: AI for Science**, Cornell University 2023
Graduate Level, *Teaching Assistant*
- **Data Structures and Algorithms**, Tsinghua University 2021
Undergraduate Level, *Teaching Assistant*
- **Introduction to Database**, Tongji University 2019
Undergraduate level, *Teaching Assistant*

Mentorship

Jiarui Liu	2025
• MS at CMU (currently CS PhD student at CMU LTI). • Mentored Jiarui on RL-finetuning LLMs for deductive reasoning.	
Junjie (Oscar) Yin	2023-2024
• Undergrad at Johns Hopkins (currently CS PhD student at UW). • Mentored Oscar on finetuning quantized LLMs.	
Jiahao Dong	2023-2024
• MS at Cornell Tech (currently ML Engineer at Distyl AI). • Mentored Jiahao on finetuning quantized LLMs.	
Jianan (Canal) Li	2023
• Undergrad at Cornell (currently CS PhD student at Berkeley). • Mentored Canal on benchmarking geometric learning methods and datasets for materials.	
Yining Huang	2023
• Undergrad at Northwestern (currently CS PhD student at UPenn). • Mentored Yining on benchmarking geometric learning methods and datasets for materials.	
Bowen Yang	2022-2023
• MS at Georgetown (currently PhD student at UAlberta). • Mentored Bowen on drug relation extraction using multi-modal data.	
Zhengchao Luo	2021-2022
• Undergrad at HUST (currently PhD student at Peking). • Mentored Zhengchao on predicting quantum chemical properties of molecules using GNNs.	